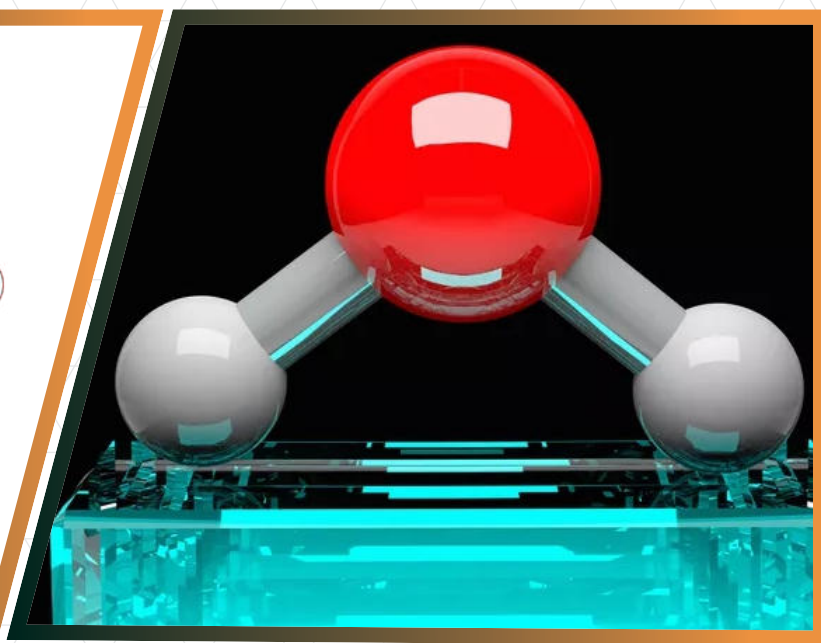
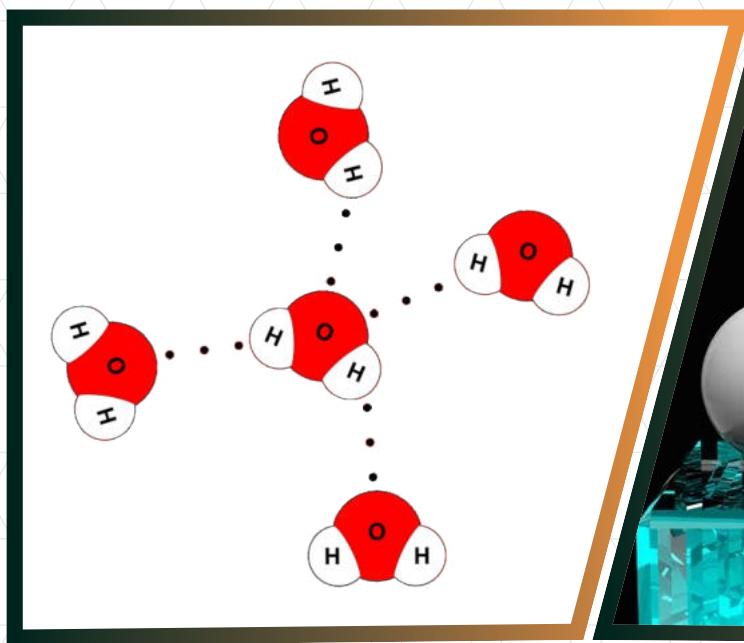




UNIT 06

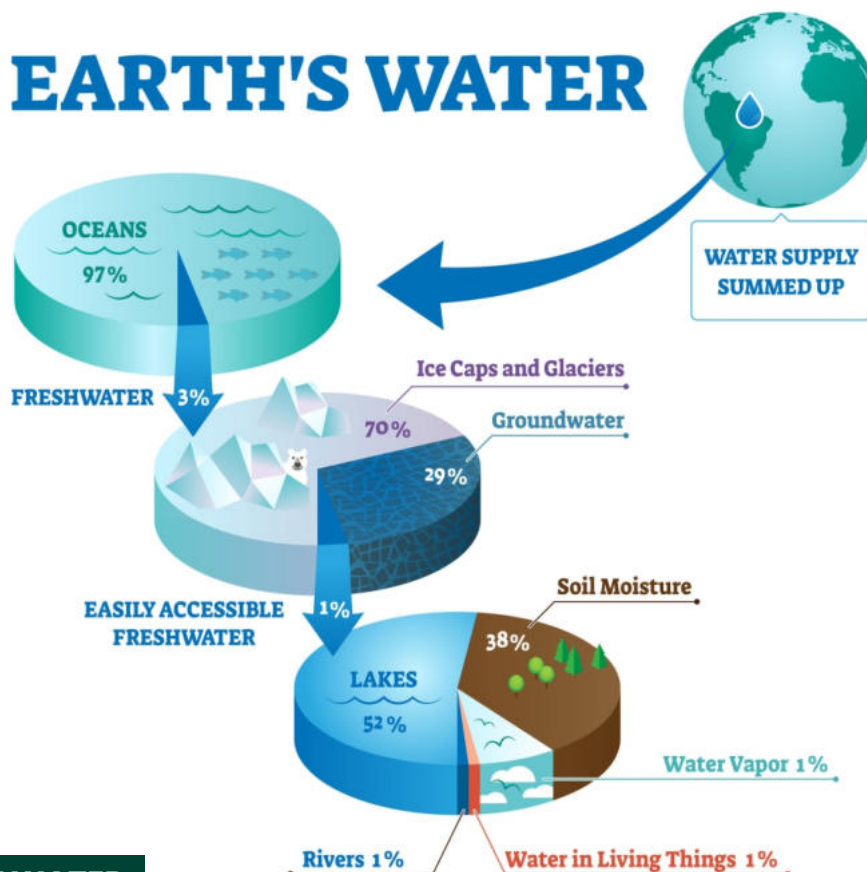
ENVIRONMENTAL CHEMISTRY-II (WATER)



WATER

OCCURRENCE OF WATER

Water is widely distributed on Earth and occurs in various forms and locations. The primary occurrence of water is in the form of vast bodies of water, such as oceans, which cover about 97% of the Earth's surface. These oceans contain saltwater, making them saline. Freshwater occurs in lakes, rivers, streams, and underground reservoirs known as aquifers. These freshwater sources are vital for drinking water, irrigation, and supporting aquatic ecosystems. Glaciers and ice caps also hold a significant amount of water, mainly in polar regions, and they gradually release water through melting.



IMPORTANCE OF WATER

1. Water is essential for hydration and maintaining the body's fluid balance.
2. It plays a crucial role in digestion, nutrient absorption, and waste removal.
3. Water is vital for regulating body temperature and preventing overheating.
4. It supports healthy cell function and lubricates joints, enabling smooth movement.
5. Water is necessary for proper brain function, concentration, and overall cognitive performance.

PROPERTIES OF WATER

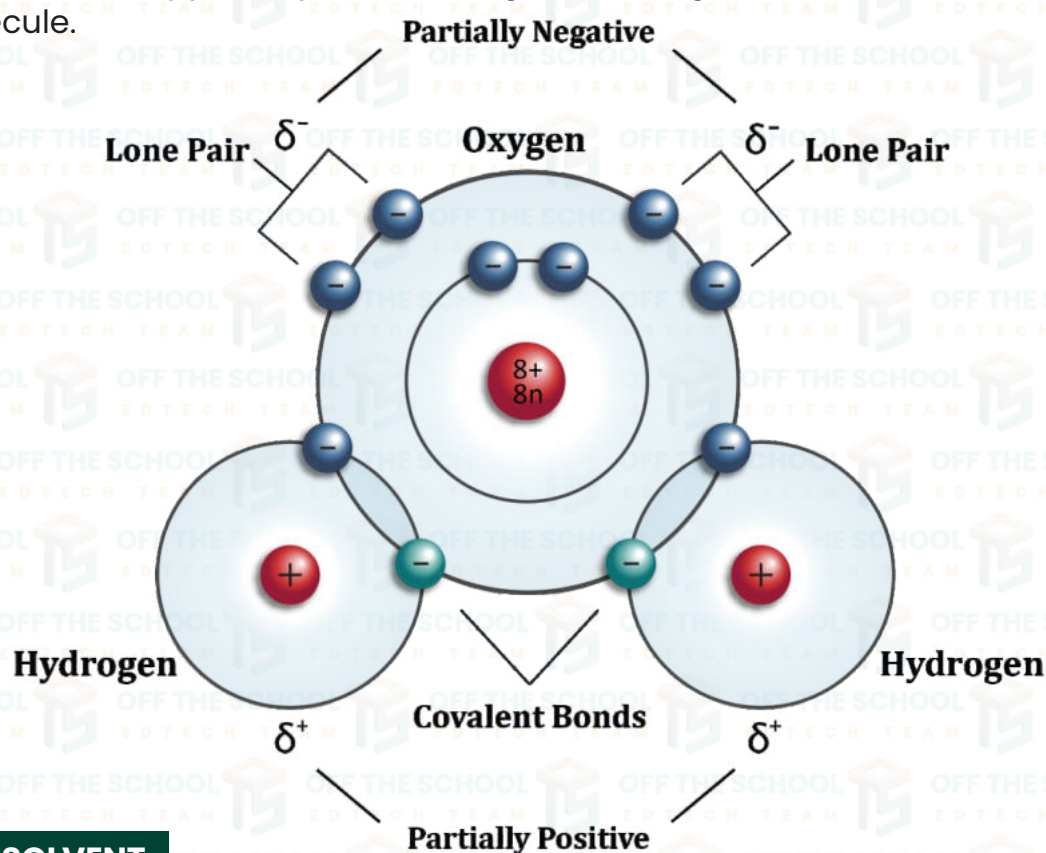
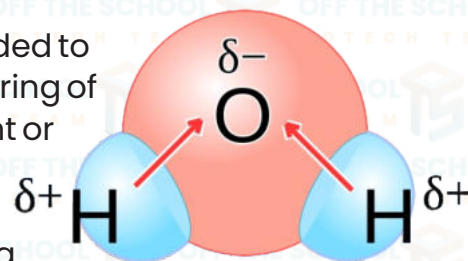
1. Pure water is odorless and tasteless.
2. It has a high heat capacity, allowing it to absorb and release heat slowly.
3. Water exhibits high surface tension, allowing it to form droplets and support certain organisms.
4. It has a high boiling point and freezing point compared to other substances.
5. Pure water is an excellent solvent, capable of dissolving many substances due to its polar nature.

COMPOSITION OF WATER

Water (H_2O) is a compound composed of two hydrogen (H) atoms bonded to one oxygen (O) atom. The ratio of hydrogen to oxygen atoms in water is 2:1. The oxygen atom forms covalent bonds with the two hydrogen atoms through sharing of electrons, resulting in a bent molecular shape. This arrangement gives water its unique properties and allows it to participate in various chemical reactions and biological processes.

STRUCTURE OF WATER

The structure of water consists of two hydrogen atoms bonded to one oxygen atom. These bonds are formed through the sharing of electrons, creating covalent bonds. The molecule has a bent or V-shaped geometry due to the arrangement of the atoms. The oxygen atom holds a partial negative charge, while the hydrogen atoms carry partial positive charges, resulting in a polar molecule.



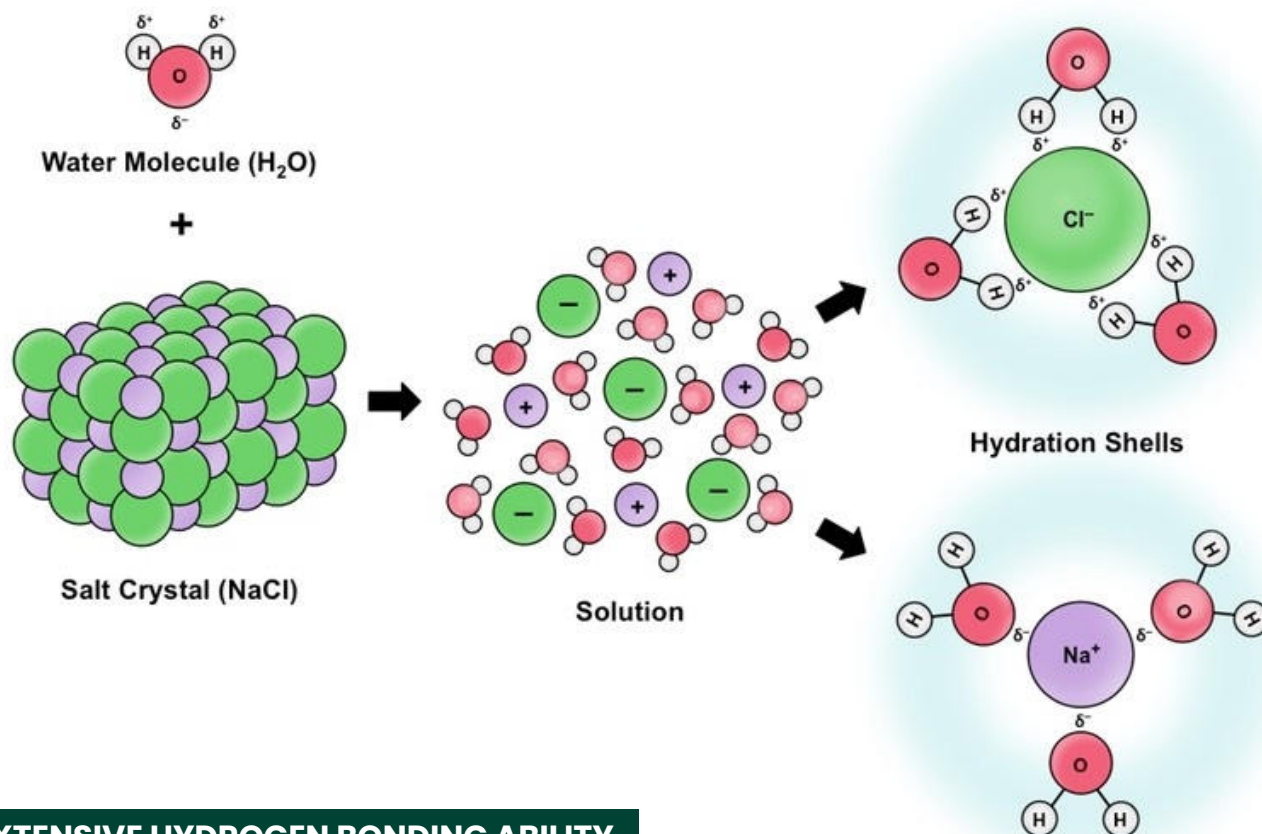
WATER AS SOLVENT

Water can dissolve practically all minerals, water is known as the universal solvent. Water's capacity to dissolve compounds is due to two distinct qualities of the molecule:

1. Polar nature of water
2. Extensive hydrogen bonding ability.

POLAR NATURE OF WATER

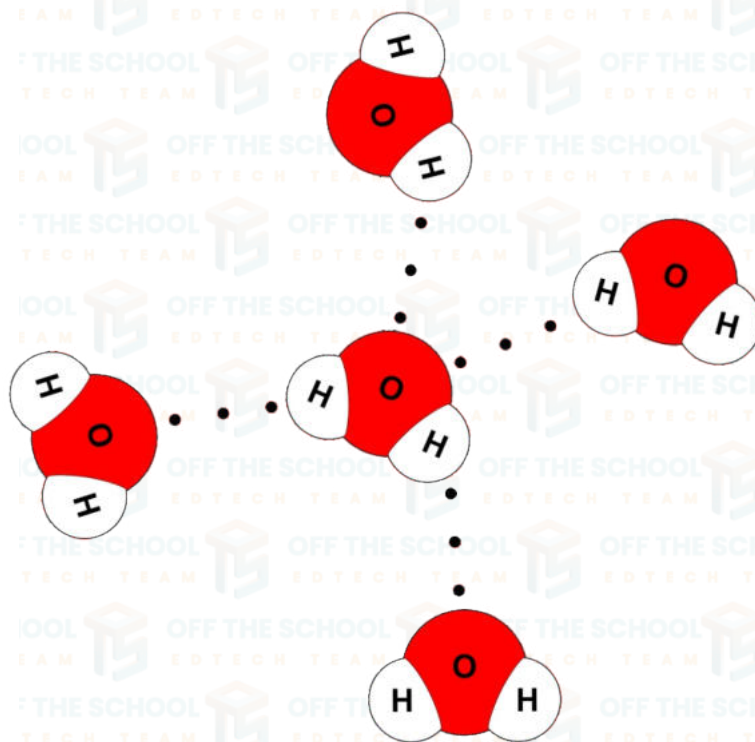
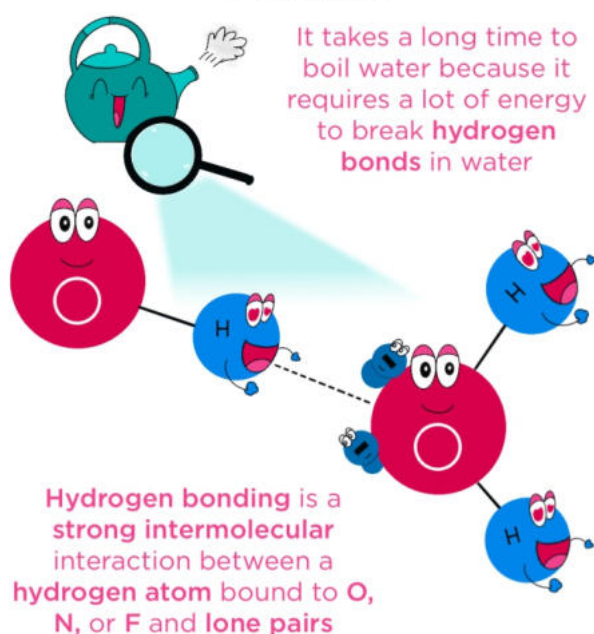
Water is a unique substance due to its polar nature, which arises from the electronegativity difference between oxygen and hydrogen atoms. In a water molecule, oxygen attracts electrons more strongly, creating a partial negative charge near the oxygen atom and partial positive charges near the hydrogen atoms. This polarity gives water distinct properties. It allows water molecules to form hydrogen bonds with one another. Water's polarity also makes it an excellent solvent for polar and ionic substances.



EXTENSIVE HYDROGEN BONDING ABILITY

The oxygen and hydrogen atoms make up the water molecule. One H_2O molecule can create hydrogen bonds with maximum four additional H_2O molecules stacked tetrahedrally around the H_2O molecule due to two O-H bonds and two lone pairs. By establishing hydrogen bonds with various polar non-ionic molecules containing hydroxyl groups ($-\text{OH}$), such as alcohols, organic acids, glucose, sugar, and so on, water is able to dissolve them.

What is Hydrogen Bonding



SOFT AND HARD WATER

SOFT WATER

It lathers with soap. Water which is obtained from the rains is soft water. This water is suitable for household purposes, for example, laundry and cleaning.

Soft water is water that contains low levels of dissolved minerals, particularly calcium and magnesium ions, resulting in reduced scaling and improved soap lathering.

Water can be classified as hard water and soft water.

HARD WATER

It is known as hard water because of the presence of calcium and magnesium salts.

Hard water does not lather with soap but instead forms a precipitate.



HARD WATER	SOFT WATER
1. It is rich in minerals.	Contains very few elements.
2. Soap is not so effective.	Soap is easily effective.
3. No foam and lather from soaps.	Bubbly lather from soaps.
4. Leaves spots on the washed dishes after they are dried.	Does not leave any spots on dishes after they are dried.
5. Contains minerals like magnesium and calcium.	Contains sodium ion.
6. Sometimes preferred drinking water.	Sometimes not preferred drinking water.
7. Example: Groundwater like deep wells.	Example: Rainwater.
8. Hair and skin become dry.	Hair and skin become soft.

TYPES OF HARDNESS OF WATER

Water hardness refers to the concentration of dissolved minerals, primarily calcium and magnesium ions, in water. The presence of these minerals can affect the performance of water in various applications, such as household use, industrial processes, and even the taste of drinking water. There are two main types of water hardness:

1. TEMPORARY HARDNESS

Temporary hardness is caused by the presence of bicarbonate minerals, mainly calcium bicarbonate ($\text{Ca}(\text{HCO}_3)_2$) and magnesium bicarbonate ($\text{Mg}(\text{HCO}_3)_2$), in water. It is called "temporary" because it can be easily removed or reduced by boiling the water or allowing it to stand exposed to the atmosphere.

2. PERMANENT HARDNESS

Permanent hardness, also known as non-carbonate hardness, is caused by the presence of dissolved sulfates, chlorides, and nitrates of calcium and magnesium in water. Unlike temporary hardness, permanent hardness cannot be removed by boiling.

METHODS OF REMOVING TEMPORARY HARDNESS

1. BOILING WATER

Boiling water is a simple method to remove temporary hardness. When water is boiled, the heat causes the bicarbonate minerals (calcium bicarbonate and magnesium bicarbonate) to decompose. This decomposition forms insoluble carbonates that precipitate out of the water. Once the water cools, the precipitates can be removed by allowing them to settle or through filtration.



2. CLARK'S METHOD

Clark's method is a chemical approach to remove temporary hardness from water. It involves adding lime ($\text{Ca}(\text{OH})_2$) or slaked lime to the water. The lime reacts with the bicarbonate ions, forming insoluble calcium carbonate (CaCO_3) precipitates. These precipitates can then be removed by settling or filtration, resulting in a reduction in temporary hardness.



REMOVAL OF PERMANENT HARDNESS

Chemicals are the only way to get rid of permanent hardness. Adding washing soda (Na_2CO_3) or sodium zeolite removes calcium (Ca^{2+}) and magnesium (Mg^{2+}) as insoluble salts.

USING WASHING SODA: Adding washing soda to the calcium and magnesium ions results in the formation of insoluble calcium and magnesium carbonates.



DISADVANTAGES OF WATER HARDNESS

Water hardness refers to the concentration of minerals, primarily calcium and magnesium, in water. While water hardness does not pose significant health risks, it can have several disadvantages and negative effects. Here are some disadvantages of water hardness:

1. Hard water can form scale deposits on pipes, faucets, and appliances.
2. Hard water makes it difficult for soap and detergents to lather and dissolve effectively.
3. Hard water can reduce the efficiency of water heaters.
4. Hard water can damage clothing and fabrics.
5. Hard water can dry out skin and hair.
6. Hard water can increase maintenance and costs.
7. Hard water can have an environmental impact.

Previous Year's Questions



How can we remove temporary hardness of water by Clark's method.

[2023]

WATER POLLUTANTS

Water pollutants can be classified into several categories based on their sources and characteristics.

Here are some common water pollutants that you can consider for a matric class:

1. INDUSTRIAL POLLUTANTS

These include various chemicals and waste materials discharged from industries into water bodies. Examples include heavy metals (lead, mercury, cadmium), organic solvents, acids, alkalis, and toxic chemicals.

2. AGRICULTURAL POLLUTANTS

Agricultural activities can contribute to water pollution through the use of fertilizers, pesticides, and herbicides. These chemicals can enter water bodies through runoff, leading to contamination. Nitrate and phosphate from fertilizers are particularly concerning as they can cause eutrophication, leading to oxygen depletion in aquatic ecosystems.

3. HOUSEHOLD WASTE

Wastewater from households and urban areas can contain organic matter, pathogens, nutrients, and various pollutants. Improperly treated or untreated sewage can introduce bacteria, viruses, and parasites into water bodies, posing health risks and degrading water quality.

EFFECTS OF WATER POLLUTANTS ON LIFE

1. Water pollutants can cause diseases such as cholera, diarrhea, and typhoid.
2. Water pollutants can harm aquatic life, such as fish, algae, and coral reefs.
3. Water pollutants can deplete ecosystems, such as wetlands and rivers.
4. Water pollutants can disturb food chains, by making fish and other animals unsafe to eat.
5. Water pollutants can make water unsafe to drink, leading to health problems such as diarrhea and kidney disease.

WATER BORNE DISEASES

Waterborne infectious illness diseases spread when people come into contact with contaminated water, either by drinking it, swimming in it, or using it to cook or bath. Following are the most common illnesses.

Previous Year's Questions



Write any six water-borne diseases.[2025]

1. CHOLERA: Cholera is an acute diarrheal illness caused by the bacterium *Vibrio cholerae*. It is spread through contaminated water or food. Symptoms of cholera include watery diarrhea, vomiting, and muscle cramps. If left untreated, cholera can be fatal.

2. TYPHOID FEVER: Typhoid fever is a serious illness caused by the bacterium *Salmonella typhi*. It is spread through contaminated food or water. Symptoms of typhoid fever include fever, headache, fatigue, cough, constipation, and rose-colored spots on the skin. If left untreated, typhoid fever can be fatal.

3. DIARRHEA: Diarrhea is a common illness that causes loose, watery stools. It can be caused by a variety of factors, including bacteria, viruses, parasites, and food poisoning. Diarrhea can be mild or severe, and it can lead to dehydration.

4. DYSENTERY: Dysentery is a type of diarrhea that is caused by bacteria or parasites. It is characterized by bloody stools and severe abdominal cramps. Dysentery can be treated with antibiotics, but it can be fatal if left untreated.

5. HEPATITIS A: Hepatitis A is a liver infection caused by the hepatitis A virus. It is spread through contaminated food or water. Symptoms of hepatitis A include fever, fatigue, nausea, vomiting, jaundice, and abdominal pain. Hepatitis A is usually a mild illness, but it can be serious in some cases.